

Customer Churn Analysis

1. Executive Summary

Customer churn poses a direct threat to recurring revenue and long-term business growth. This project establishes an end-to-end data processing and analytics framework designed to centralize customer data, compute core churn metrics, and prepare datasets for executive dashboarding in Power BI.

By leveraging SQL Server Management Studio (SSMS) to clean raw staging data and execute metric-driven SQL queries, this phase establishes a reliable data pipeline to track state-level retention, churn categories, and customer demographics.

2. Dataset Overview

- Kaggle dataset
- Entries: 6,419 records

3. Key Components:

- **Staging Environment:** Raw transactional and demographic data is ingested into the SQL database table `stg_Churn`.
- **SQL Transformation Layer:** Queries run directly in SQL Server to compute proportions, total counts, and percentage contributions across multiple dimensions.
- **Power BI Connection:** Cleaned datasets and views are pulled into Power BI for dashboard development.

4. Core Metrics & SQL Analysis Findings

During the SQL analysis phase, several key performance indicators (KPIs) and demographic breakdowns were calculated:

a) Overall Customer & Churn Overview

- **Customer Status Distribution:** SQL queries calculate total customer counts and percentage contributions grouped by customer status (e.g., Joined, Stayed, Churned).
- **Revenue Contribution:** Evaluates revenue generated versus lost revenue from churned users to highlight financial exposure.

b) Geographic Breakdown (State-Level Metrics)

State-by-state analysis was conducted to pinpoint geographic concentrations of total accounts and churn percentage:

- **High-Density States:** States like Uttar Pradesh, Tamil Nadu, and Maharashtra represent the largest customer concentrations.

c) Primary Drivers & Category Explorations

SQL queries were structured to analyse distribution across key operational fields:

- **Contract Types:** Evaluating churn rates across Month-to-Month, 1-Year, and 2-Year plans.
- **Payment Methods:** Mapping churn distribution against manual versus automated payment methods.
- **Churn Categories & Reasons:** Identifying core categories driving service cancellations (e.g., Competitor offers, price points, service quality).

5. Key Takeaways & Initial Findings

- **Geographic Concentration:** A significant portion of the customer base resides in key regional hubs, requiring targeted local retention strategies.
- **Data Cleanliness:** Running preliminary aggregations in SQL Server ensures that data inconsistencies are resolved before creating visual reports in Power BI.
- **Pipeline Efficiency:** Preparing data views and aggregated queries in SQL optimizes Power BI performance by delegating heavy calculations to the database server.

6. Data Analysis using SQL

i. Total churn by Gender:

	Gender	TotalCount	Percentage
1	Male	2370	36.93
2	Female	4048	63.07

ii. Churn Based on Contract:

	Contract	TotalCount	Percentage
1	Month-to-Month	3286	0.51
2	One Year	1413	0.22
3	Two Year	1719	0.27

iii. Total Revenue by Customer Status:

	Customer_Status	TotalCount	TotalRev	RevPercentage
1	Joined	411	49281.5598697662	0.0025
2	Churned	1732	3411960.5796299	0.1752
3	Stayed	4275	16010148.2622757	0.8222

iv. Churn by State:

	State	TotalCount	Percentage
2	Tamil Nadu	600	9.35
3	Maharashtra	504	7.85
4	Karnataka	470	7.32
5	Haryana	398	6.20
6	Andhra Pradesh	395	6.15
7	West Bengal	368	5.73
8	Punjab	342	5.33
9	Bihar	336	5.24
10	Gujarat	335	5.22
11	Jammu & Kashmir	320	4.99
12	Madhya Pradesh	288	4.49
13	Telangana	281	4.38
14	Rajasthan	259	4.04
15	Kerala	200	3.12
16	Odisha	152	2.37
17	Assam	139	2.17

v. Null Value Check:

	_Count	Total_Long_Distance_Charges_Null_Count	Total_Revenue_Null_Count	Customer_Status_Null_Count	Churn_Category_Null_Count	Churn_Reason_Null_Count
1		0	0	0	4686	4686

vi. Replace Null Values:

	Customer_ID	Gender	Age	Married	State	Number_of_Referrals	Tenure_in_Months	Value_Deal	Phone_Service	Multiple_Lines	Inter
1	11098-MAD	Female	30	Yes	Madhya Pradesh	0	31	Deal 1	Yes	No	Yes
2	11114-PUN	Male	51	No	Punjab	5	9	Deal 5	Yes	No	Yes
3	11167-WES	Female	43	Yes	West Bengal	3	28	Deal 1	Yes	Yes	Yes
4	11179-MAH	Male	35	No	Maharashtra	10	12	None	Yes	No	Yes
5	11180-TAM	Male	75	Yes	Tamil Nadu	12	27	Deal 2	Yes	No	Yes
6	11241-MAD	Female	41	Yes	Madhya Pradesh	4	11	None	Yes	No	Yes
7	11244-JAM	Female	20	No	Jammu & Kashmir	3	9	None	Yes	Yes	Yes
8	11251-UTT	Female	51	No	Uttarakhand	1	19	None	Yes	Yes	No
9	11262-HAR	Female	73	Yes	Haryana	5	32	None	Yes	Yes	Yes
10	11263-HAR	Female	41	No	Haryana	13	31	Deal 2	Yes	Yes	Yes
11	11264-MAH	Female	27	Yes	Maharashtra	14	17	Deal 5	No	No	Yes
12	11272-UTT	Female	65	No	Uttar Pradesh	0	19	Deal 5	Yes	Yes	Yes
13	11277-UTT	Male	66	Yes	Uttar Pradesh	10	23	None	Yes	No	Yes
14	11288-MAD	Male	52	No	Madhya Pradesh	6	24	None	Yes	No	No
15	11290-JAM	Female	70	Yes	Jammu & Kashmir	0	36	Deal 5	Yes	Yes	Yes

7. Dashboard in Power BI



8. Recommendations

- **Incentivize Long-Term Contracts:** Transition month-to-month subscribers to annual plans using targeted discounts to counter the high ~46% churn rate on short-term commitments.
- **Implement Competitor Win-Back Strategies:** Address the leading churn driver ("Competitor") by running price-matching programs, enhancing core features, and highlighting product differentiation.
- **Automate Auto-Pay Adoption:** Reduce payment-friction churn by offering small billing incentives for switching to automatic credit card or bank payments instead of manual checks.